

# Experiment Report

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## Abstract

This report examines two classification techniques, K-Nearest Neighbors (KNN) and Nearest Centroid Classification, using Scikit-learn. The study focuses on understanding the decision boundaries, the impact of hyperparameters like weights, k value (for KNN) and shrinkage (for Nearest Centroid). Both classifiers are trained on the iris dataset, and the effects of parameter tuning on model performance are visualized using decision boundary plots. The results show that adjusting these parameters can significantly influence the decision boundaries and classification accuracy.

## Introduction

Classification is a fundamental task in machine learning, and two common techniques are K-Nearest Neighbors (KNN) and Nearest Centroid Classification. Both methods aim to classify data points based on their proximity to predefined class centroids or neighbors. This report investigates the decision boundaries created by these methods using the iris dataset. The objective is to compare the effects of different parameter settings, such as distance-based weighting in KNN and shrinkage in Nearest Centroid Classification, to better understand how these parameters impact the classification performance.

**K-Nearest Neighbors Classification (KNN):** This method identifies the 'k' nearest data points to a query point and assigns a class based on the majority vote of these neighbors. The influence of each neighbor can be weighted by its distance from the query point.

**Nearest Centroid Classification:** This technique classifies a point based on its nearest class centroid. Centroids are calculated as the mean position of the training points for each class. A shrinkage threshold can be applied to reduce the influence of outliers.

## Materials and Methods

**Dataset:**

The iris dataset is used in both classification methods. This dataset consists of 150 samples from three species of iris flowers, each with four features: sepal length, sepal width, petal length, and petal width. For simplicity, only two features (sepal length and sepal width) are used for visualization.

**K-Nearest Neighbors Classification:**

- The `KNeighborsClassifier` from Scikit-learn was used. The k-value was tuned through experimentation to find the optimal value.
- The dataset was split into training and testing subsets using `train_test_split`.

- A Pipeline was created to scale the features using StandardScaler and then apply the KNN model.
- Decision boundaries were plotted for two different weighting schemes: "uniform" and "distance."

Nearest Centroid Classification:

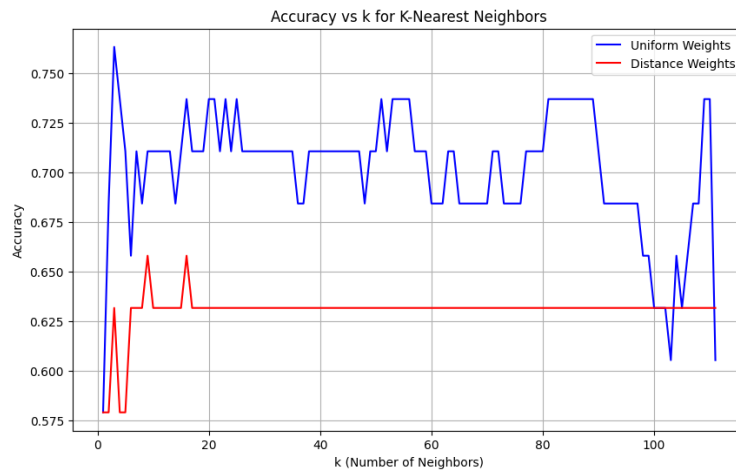
- The NearestCentroid classifier was used. The shrinkage threshold (shrink\_threshold) was adjusted during parameter tuning to identify the optimal value, with visualizations provided for both cases with and without shrinkage.
- Decision boundaries were visualized for both cases, and classification accuracy was compared.
- Color maps were used to highlight the different classes and decision regions.

Parameter Tuning:

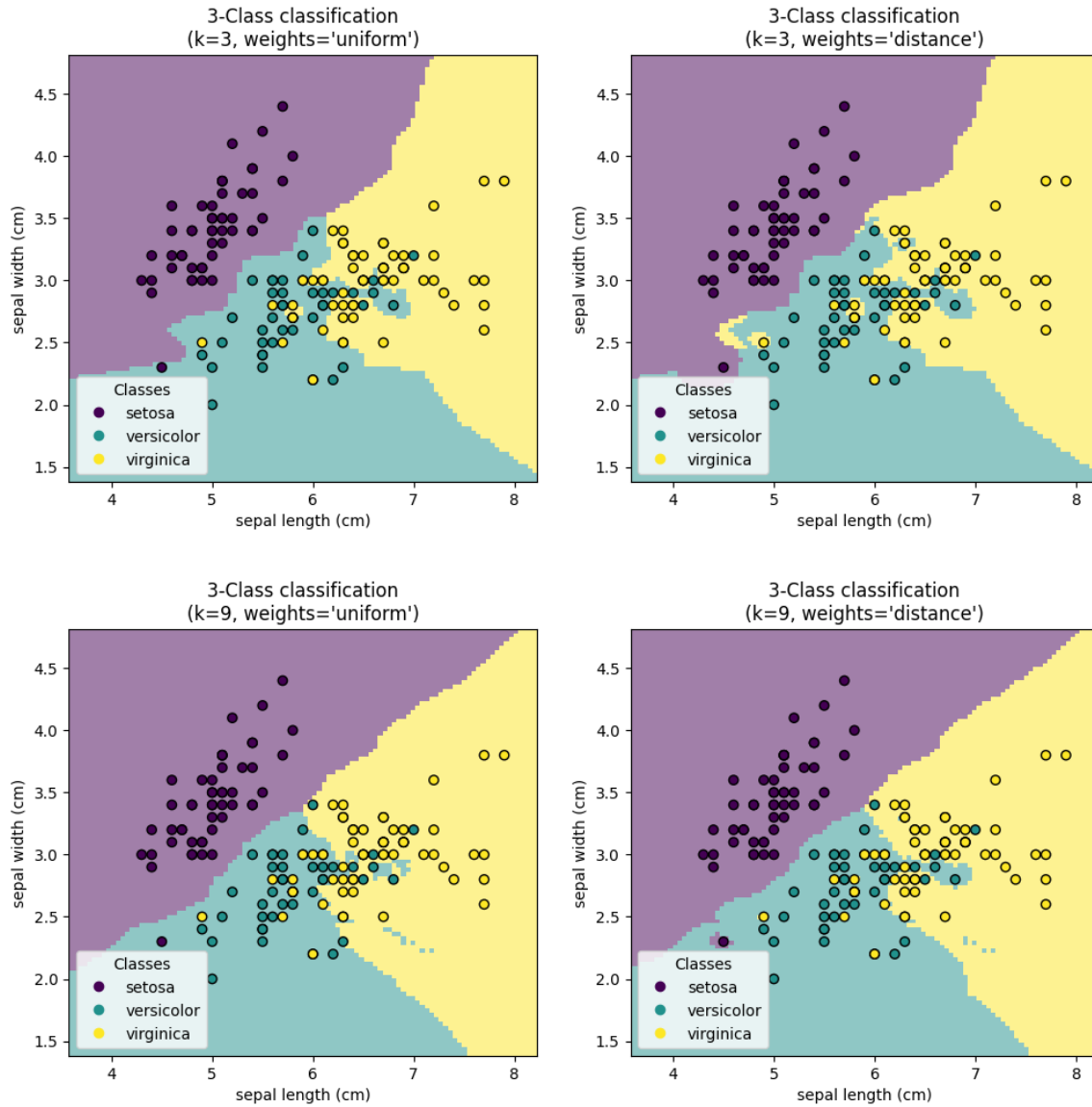
Throughout the experiment, both the  $k$ -value in K-Nearest Neighbors and the shrinkage threshold in Nearest Centroid Classification were adjusted to find the combination that resulted in the highest accuracy.

## Results

K-Nearest Neighbors Classification:



*Figure 1.* The plot shows the accuracy of the K-Nearest Neighbors (KNN) classifier versus the number of neighbors  $k$  for uniform and distance weighting schemes. The x-axis represents the number of neighbors, while the y-axis shows accuracy. The blue line indicates uniform weights, and the red line indicates distance weights. The plot demonstrates how accuracy changes with  $k$  for both weighting methods.



*Figure 2.* The plot shows the decision boundaries of a K-Nearest Neighbors (KNN) classifier using uniform and distance weights. The left plot illustrates the boundaries with uniform weights, and the right plot shows the boundaries with distance weights. In each plot, shaded regions represent predicted classes, and scatter points indicate test samples colored by their true class labels. The upper plots are for  $k=3$  which accuracy is highest for uniform weighting, while the lower plots are for  $k=9$  which accuracy is highest for distance weighting. These plots highlight how different weighting schemes and  $k$  values influence the classifier's decision regions.

The numerical analysis reveals that uniform weighting consistently outperforms distance weighting across all tested values of  $k$ . The highest accuracy achieved with uniform weights is 0.7632 at  $k=3$ , whereas the maximum accuracy with distance weights is 0.6579 at  $k=9$ . This

indicates that, for this dataset, the uniform weighting scheme provides superior classification performance compared to the distance weighting scheme.

### Nearest Centroid Classification:

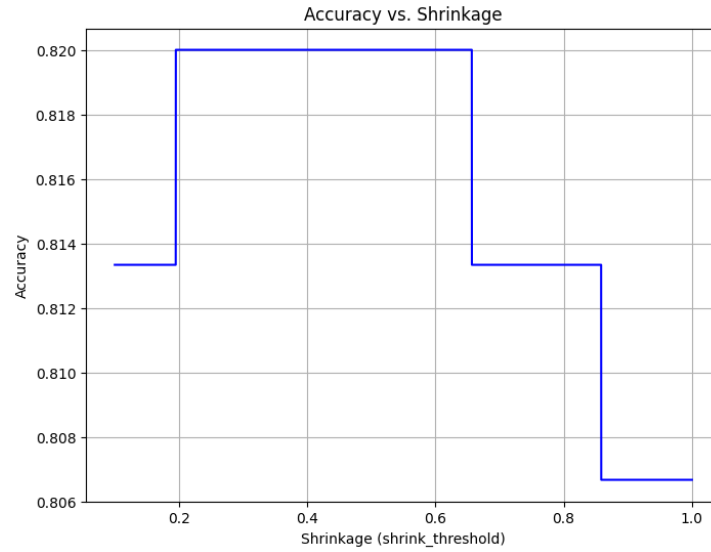


Figure 3. The plot shows how accuracy varies with different shrinkage values. The x-axis represents the shrinkage (*shrink\_threshold*) from 0 to 1, and the y-axis shows the accuracy. The blue line indicates accuracy across these shrinkage values, highlighting how the model's performance changes with different levels of shrinkage.

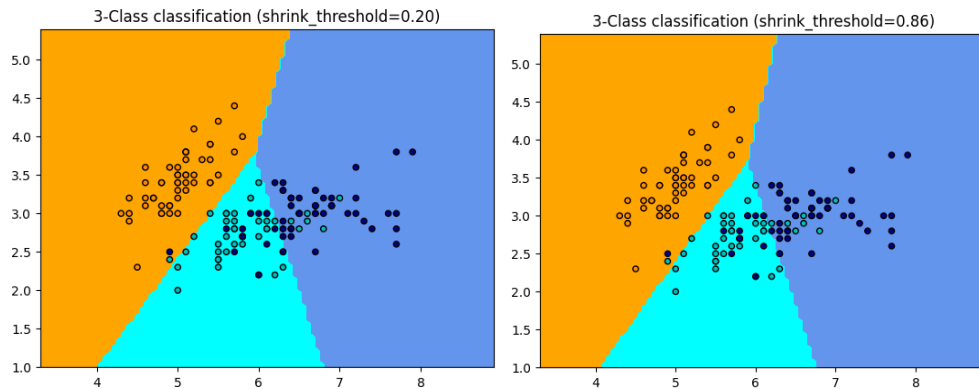


Figure 4. The left plot shows the decision boundary for the *shrink\_threshold* of 0.20, which resulted in the highest accuracy. The right plot shows the boundary for the *shrink\_threshold* of 0.86, which resulted in the lowest accuracy. Each plot displays the class regions in the background and the actual training data points, with titles indicating the respective *shrink\_threshold* values.

The decision boundary with a *shrink\_threshold* of 0.20 provides the best classification performance, while the boundary with a *shrink\_threshold* of 0.86 shows the poorest performance, as indicated by the accuracy of the classifier with these values.

## Discussion

### K-Nearest Neighbors Classification:

This experiment investigates the impact of different weighting schemes (uniform weights and distance weights) in the K-Nearest Neighbors (KNN) algorithm on classification accuracy, and compares their performance at different k-values. As seen in *Figure 1*, the accuracy of uniform weights fluctuates significantly with changes in k-values, reaching a peak of approximately 0.75 when k is small. However, as k increases, the accuracy of uniform weights gradually decreases. In contrast, the accuracy of distance weights remains relatively stable, hovering around 0.625 regardless of the k-value. Although distance weights show more stable performance, their overall accuracy is lower than the best performance achieved by uniform weights.

These results indicate that when using uniform weights, all neighbors have the same influence on the decision-making process. While this scheme is sensitive to changes in k, it can provide higher accuracy when k is properly adjusted. Conversely, when using distance weights, the influence of neighbors is inversely proportional to their distance from the query point, meaning that closer neighbors have a greater impact on the decision. Despite offering greater stability, this scheme generally yields lower accuracy.

The differences in performance between these weighting schemes are mainly related to how neighbors influence the classification decision. In the uniform weighting scheme, all neighbors, regardless of their distance, have equal influence. This may result in higher accuracy when k is small, as the model relies on a few close neighbors. However, as k increases, the influence of distant neighbors also increases, potentially introducing noise and reducing accuracy. In contrast, the distance weighting scheme gives more weight to closer neighbors, minimizing the interference of distant neighbors and leading to more stable performance. However, by ignoring information from distant neighbors, the overall accuracy tends to be lower than that of uniform weights. The characteristics of the dataset's class distribution may also affect the performance of the two schemes: uniform weights perform better when class boundaries are clear, whereas distance weights provide more stable results when the boundaries are less distinct.

### Nearest Centroid Classification:

As seen in *Figure 3*, it presents the relationship between the shrinkage parameter (*shrink\_threshold*) and classification accuracy in the Nearest Centroid algorithm. Shrinkage helps reduce the impact of small, less important centroids in the classification decision-making process. As observed from the plot, the accuracy fluctuates with different shrinkage values, reaching a peak of approximately 0.82 when the shrinkage value lies between 0.2 and 0.6.

The accuracy begins at around 0.814 without shrinkage, briefly increases to a higher value of 0.82 for smaller shrinkage values (around 0.2), then maintains this level before decreasing slightly as

the shrinkage parameter approaches 1.0. The overall trend indicates that shrinkage can marginally improve accuracy up to a certain threshold but tends to reduce performance when it is overly large.

This behavior suggests that introducing moderate shrinkage allows the model to generalize better by ignoring noisy or less important centroids. However, excessive shrinkage removes critical decision boundaries, leading to a decline in accuracy. Thus, the optimal shrinkage range appears to be around 0.2 to 0.6 for the given dataset.

## Conclusion

From this experiment, we can conclude the following key points:

K-Nearest Neighbors (KNN) Classification:

- The results indicate that uniform weights generally outperform distance weights in KNN classification. This suggests that, with proper adjustment of the k-value, uniform weights can achieve higher classification accuracy.
- The choice of k-value is crucial for KNN performance and should be tuned according to the characteristics of the dataset to find the optimal value.

Nearest Centroid Classification:

- In the Nearest Centroid Classification experiment, the use of shrinkage can improve classification accuracy within a certain range. Accuracy peaks when the shrinkage parameter is between 0.2 and 0.6.
- However, once the shrinkage value exceeds this range, accuracy tends to decrease. This indicates that excessive shrinkage can reduce the model's classification performance.

These conclusions provide valuable insights into the impact of different weighting schemes and parameter adjustments on classification performance, aiding in the effective selection of appropriate parameters and methods in practical applications.

## References

1. Scikit-learn. (n.d.). Classification with nearest neighbors. Retrieved from [https://scikit-learn.org/stable/auto\\_examples/neighbors/plot\\_classification.html](https://scikit-learn.org/stable/auto_examples/neighbors/plot_classification.html)
2. Scikit-learn. (n.d.). Nearest centroid classification. Retrieved from [https://scikit-learn.org/stable/auto\\_examples/neighbors/plot\\_nearest\\_centroid.html](https://scikit-learn.org/stable/auto_examples/neighbors/plot_nearest_centroid.html)